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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Database Systems (course)

Course outline

About NPTEL ()**Introduction to
Database Systems ()****Week 0 ()****Week 1 ()****Week 2 ()**☐ Introduction to ER
Model (unit?)

Week 2 : Assignment 2

The due date for submitting this assignment has passed.**Due on 2026-02-04, 23:59 IST.****As per our records you have not submitted this assignment.**

1) An entity type may have more than one attribute.

0.5 points☐ True☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

True

2) A relationship type may have more than one attribute.

0.5 points

unit=24&lesson=25)

- ☐ Entities and Relationships (unit? unit=24&lesson=26)
- ☐ Modelling Weak Entities and Design Choices (unit? unit=24&lesson=27)
- ☐ Lecture Slides (unit? unit=24&lesson=28)
- ☐ Week 2 Feedback Form: Introduction to Database Systems!! (unit? unit=24&lesson=29)
- ☐ **Quiz: Week 2 : Assignment 2 (assessment? name=191)**

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

☐ True

☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

True

3) A strong entity type always has a key.

0.5 points

☐ True

☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

True

4) A weak entity type always has a key.

0.5 points

☐ True

☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

False

5) Suppose *Total-Salary* attribute of employee entity type is computed using attributes *Base-Salary*, *HRA*, and *Other-Allowances*. Choose the most appropriate type for attribute *Total-salary*.

2 points

☐ Single valued

☐ Multi valued

☐ Composite

☐ Single-valued and derived

No, the answer is incorrect.

[Week 10 \(\)](#)[Week 11 \(\)](#)[Week 12 \(\)](#)[Download Videos \(\)](#)[Lecture Notes \(\)](#)[Problem Solving
Session - Jan 2026 \(\)](#)

Score: 0

Accepted Answers:

Single-valued and derived

6) Suppose **WorksFor** is a relationship type with two participating entity types **Employee** and **Company**. Say, it represents the full-time job of the employees. What is the appropriate cardinality ratio for “Employee: Company”?

2 points☐ 1:1☐ N:1☐ 1:N☐ M:N

No, the answer is incorrect.

Score: 0

Accepted Answers:

N:1

7) Suppose **BankAccount** is a relationship type with two participating entity types **Person** and **Account** in a bank. What is/are the **possible** cardinality ratio(s) for “Person: Account”?

2 points

P: 1-to-1; Q: 1-to-N; R: N-to-1; S: M-to-N

☐ Only P☐ Only P & Q☐ Only P, Q & R☐ All cardinality ratios are possible.

No, the answer is incorrect.

Score: 0

Accepted Answers:

All cardinality ratios are possible.

8) Suppose **Assigned** is a many-to-one relationship type with two participating entity types **Employee** and **Department**. A maximum of seven employees are assigned to a department. If in the ER diagram, the line that connects **Employee** to the relationship type **Assigned** is to be labeled with (x,y) using min-max notation, choose the correct option:

2 points☐ x = 1, y = 7

- ☐ $x = 1, y = 1$
- ☐ $x = 7, y = 1$
- ☐ $x = 7, y = 7$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x = 1, y = 1$

9) Consider the statements given below:

2 points

S1: The owner of a weak entity is always a strong entity.

S2: Entity sets must be disjoint.

Choose correct option:

- ☐ S1: True ; S2: True
- ☐ S1: True; S2: False
- ☐ S1: False; S2: True
- ☐ S1: False; S2: False

No, the answer is incorrect.

Score: 0

Accepted Answers:

S1: False; S2: False

10) Consider the following sets:

2 points

S1: {p: strong entity; q: weak entity; r: relationship; s: multi-valued attributes}

S2: {w: double-lined rectangles; x: diamond; y: rectangle ; z: double-lined ellipses}

Choose the correct matching between S1 and S2:

- ☐ p: w; q: x; r: y; s: z

- ☐ p: w; q: x; r: z; s: y
- ☐ p: y; q: w; r: x; s: z
- ☐ p: y; q: w; r: x; s: x

No, the answer is incorrect.

Score: 0

Accepted Answers:

p: y; q: w; r: x; s: z

11) Consider the following attributes of the various entity types

2 points

A1: Customer-Mobile-Number(Domain: Banking)

A2: Customer-City-Name(Domain: Banking)

A3: Airport-Code(Domain: Airline)

A4: Student-Email-Id(Domain:University)

Consider the typical modeling situations about the given domains and choose the **FALSE** statement:

- ☐ Attribute A1 may be a key attribute of customer entity
- ☐ Attribute A2 may be a key attribute of customer entity
- ☐ Attributes A3 may be a key attribute of airport entity
- ☐ Attribute A4 may be a key attribute of student entity

No, the answer is incorrect.

Score: 0

Accepted Answers:

Attribute A2 may be a key attribute of customer entity

12) Consider two entity sets $A = \{a,b,c,d\}$ and $B = \{w,x,y,z\}$ and they participate in a relationship R. The instances of R are given as: $\{(a,w),(a,x),(c,y),(d,z),(c,z)\}$. Which one of the following is correct: **2 points**

- ☐ Cardinality ratio of A:B is one-to-many; A participates partially; B participates completely
- ☐ Cardinality ratio of A:B is many-to-one; A participates completely; B participates completely
- ☐ Cardinality ratio of A:B is one-to-many; A participates completely; B participates partially
- ☐ Cardinality ratio of A:B is many-to-many; A participates partially; B participates partially

No, the answer is incorrect.

Score: 0

Accepted Answers:

Cardinality ratio of A:B is one-to-many; A participates partially; B participates completely

13) Consider two entity sets $A = \{a, b, c, d, e\}$ and $B = \{w, x, y, z\}$ and consider a relation R as a many-to-one relationship from entity set A to entity set B. Assume that A and B participate totally in R. Which one of the following is TRUE about R: **2 points**

- ☐ Every entity in A is associated with exactly one entity in B
- ☐ Some entity in A is associated with more than one entity in B
- ☐ Every entity in B is associated with exactly one entity in A
- ☐ Every entity in B is associated with at most one entity in A

No, the answer is incorrect.

Score: 0

Accepted Answers:

Every entity in A is associated with exactly one entity in B